



**HKU
Med**

School of Clinical Medicine
Department of Medicine
香港大學內科學系

Pleural Effusion in a chronic smoker:

Lung Cancer

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Department of Medicine
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Learning Objectives :

- (1) Learn the pathophysiology, common causes and investigations of **pleural effusion**
- (2) Understand the pathology and natural history of **lung cancer**, and **cigarette smoking** as the predominant cause
- (3) Understand the **symptoms and signs** of **lung cancer** with reference to the anatomy of the thoracic structures and cancer metastasis
- (4) Learn the **principles** of investigations, staging and treatment of **lung cancer**



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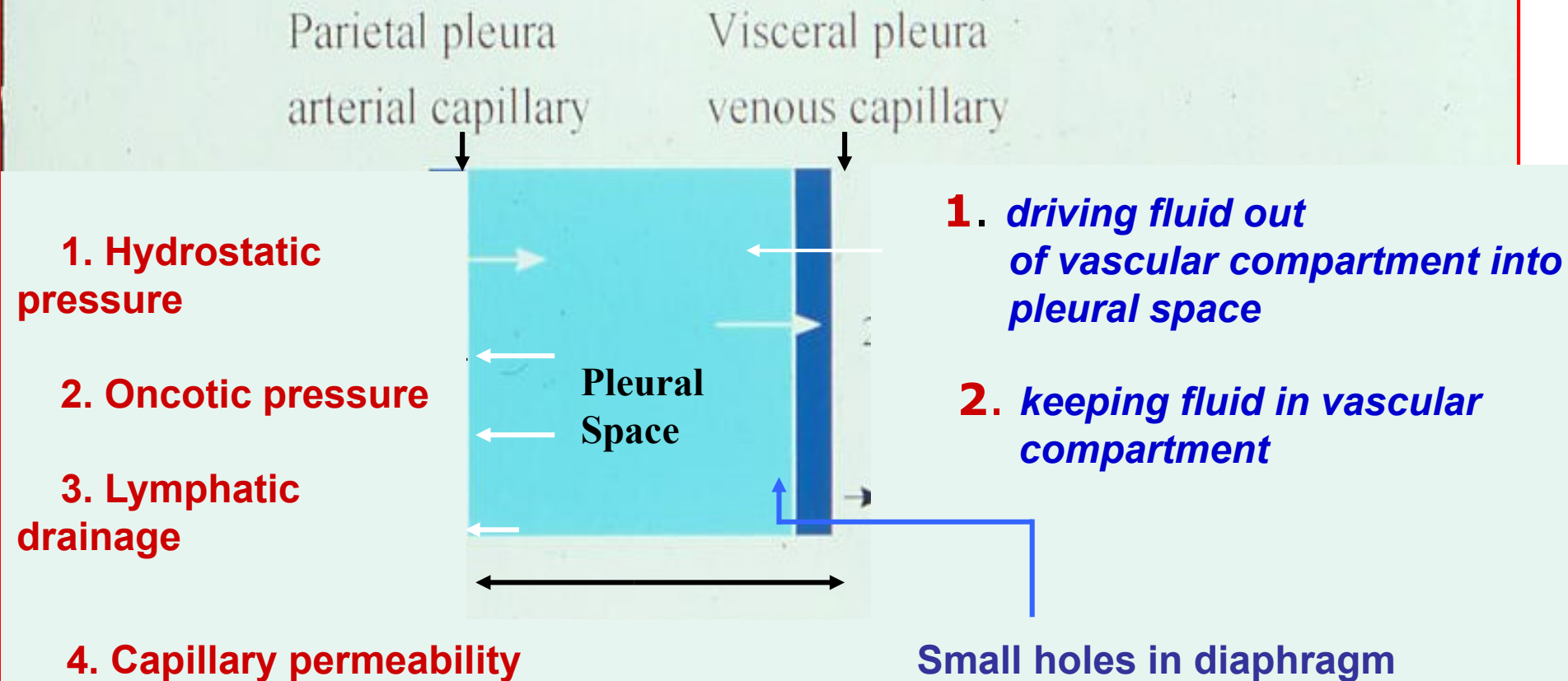
**M/64, chronic smoker,
retired restaurant waiter**

- **Diagnostic + therapeutic tapping x 2**
- **Exudative changes**
- **Atypical cells with large N/C ratio.
Need to rule out malignancy**
- **Non-diagnostic pleural fluid cytology
and percutaneous biopsy**



PATHOPHYSIOLOGY OF PLEURAL EFFUSION

Four Factors :





INCREASE IN HYDROSTATIC PRESSURE

- Congestive heart failure
 - Renal failure (fluid retention)
 - Other forms of volume overload
-

DECREASE IN ONCOTIC PRESSURE (↓ albumin)

- Cirrhosis
 - Nephrotic syndrome
 - Severe malnutrition
 - Other forms of hypoalbuminemia
-

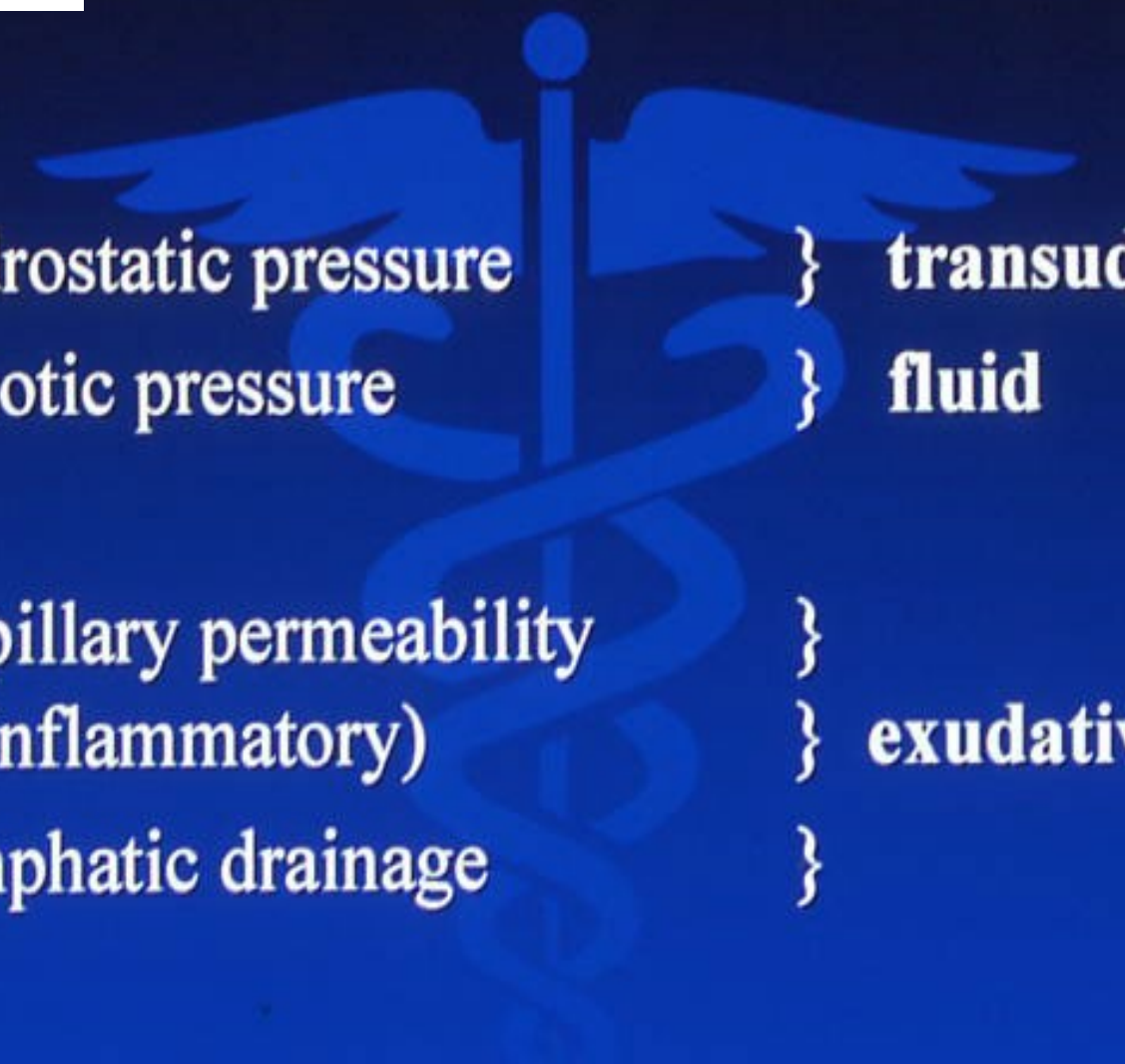


PLEURAL INFLAMMATION : INCREASE IN CAPILLARY PERMEABILITY

- infection of lung or pleura (including TB)
 - collagen vascular disease (involving pleura)
 - malignant involvement of pleura
 - intra-abdominal process (e.g. pancreatitis, liver abscess)
-

DECREASE IN LYMPHATIC DRAINAGE

- malignant infiltration of the mediastinal, pulmonary or pleural lymphatics
 - necrotising or granulomatous infections
-

- 
- $\uparrow$ hydrostatic pressure
 - $\downarrow$ oncotic pressure
- } **transudative fluid**
-
- $\uparrow$ capillary permeability
(inflammatory)
 - $\downarrow$ lymphatic drainage
- } **exudative fluid**

INVESTIGATIONS (1)

- Diagnostic thoracentesis and pleural fluid analysis
 - *appearance*
 - *chemistry* . protein & LDH -
 - (1) pleural fluid protein / serum protein > 0.5
 - (2) pleural fluid LDH / serum LDH > 0.6
 - (3) pleural fluid LDH $> 2/3$ normal upper limit for serum
 - exudative if ≥ 1 criterion +ve
 - glucose & pH - ↓ together in certain exudative process, e.g. empyema, rheumatoid pleurisy
 - *cell count & differential*
 - *cytology*
 - *microbiology*

INVESTIGATIONS (2)

- percutaneous pleural biopsy - exudative effusion
especially to confirm TB or cancer
 - thoracoscopy / pleuroscopy
 - identification of primary causes
-



Carcinoma of Lung

- **Epidemiology & Causative Factors**
- **Pathology**
- **Clinical Features**
- **Investigations & Staging**
- **Management**
- **Prognosis**



In Hong Kong

Non-small cell lung cancer → 85% of lung cancer

Male patients. 80% are smokers

Female patients . < 40% are smokers

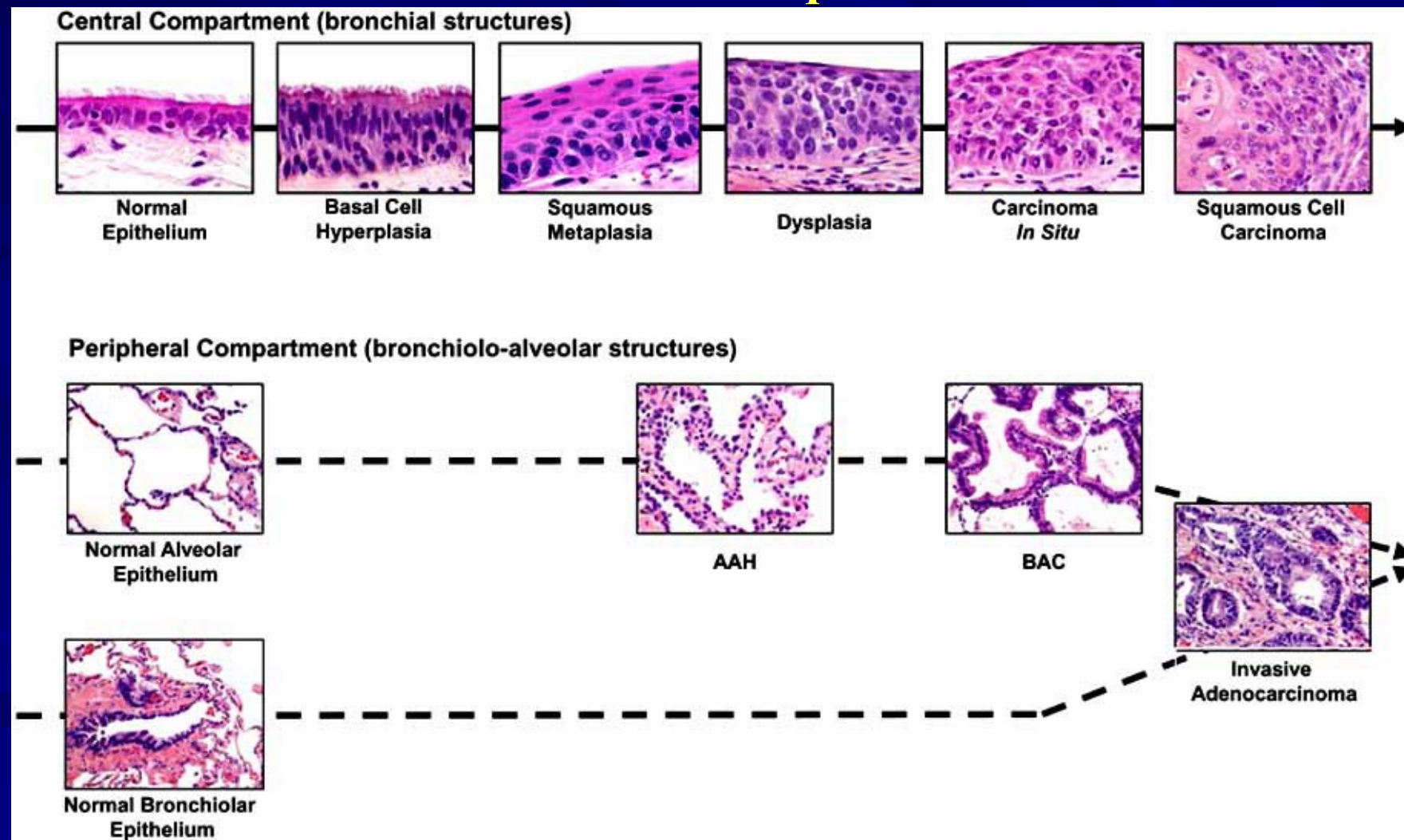
- . > 75% adenocarcinoma**

- . Risk factors**

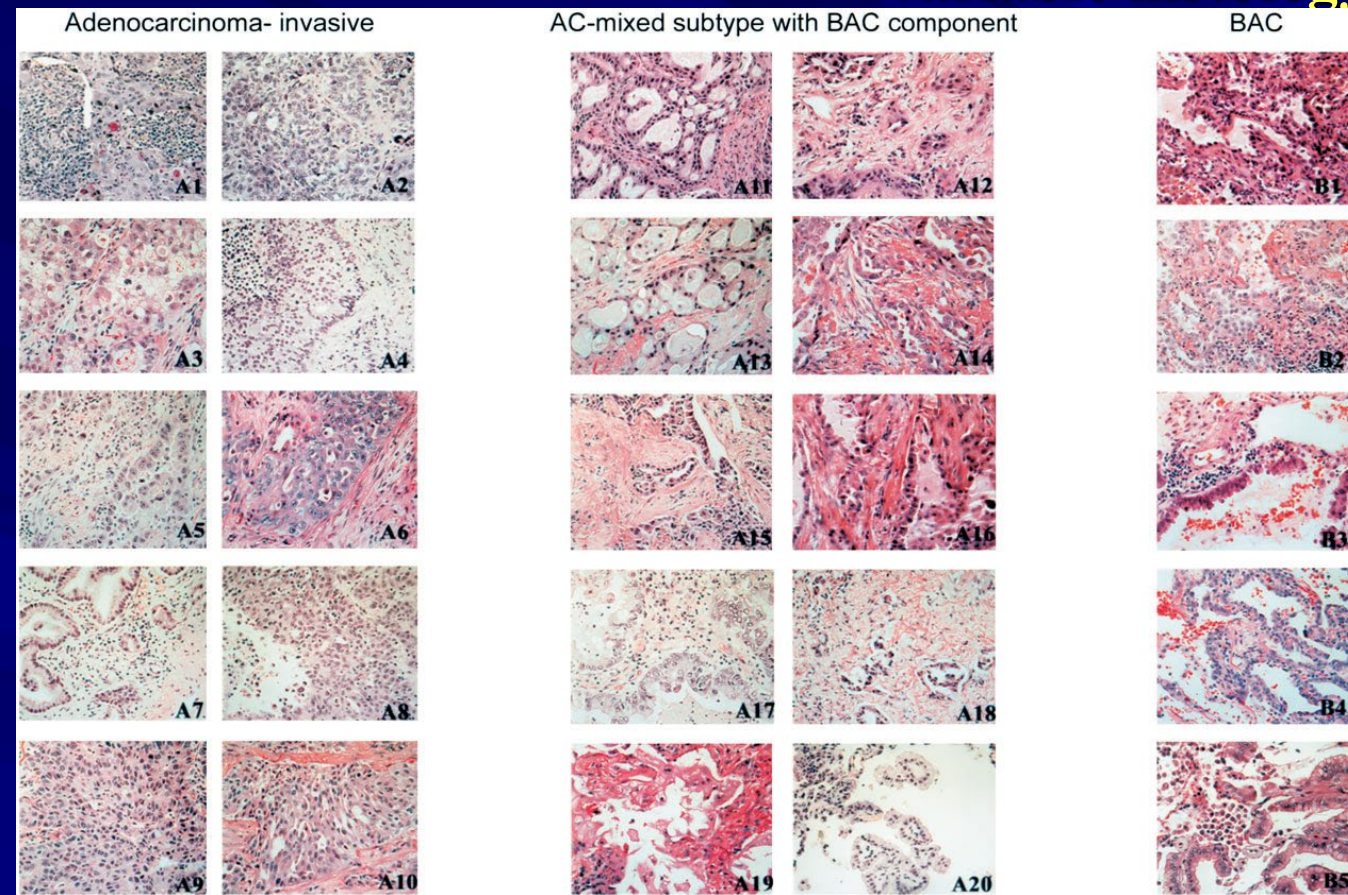
- (a) Environmental : Passive smoking, diet, cooking fumes**

- (b) Genetic *EGFR* gene mutation important in some non-smoking patients with adenocarcinoma**

Pathogenesis of adenocarcinoma and squamous cell carcinoma



Pulmonary Adenocarcinoma: variable histology





WHO classification of lung tumor histology

I. Malignant epithelial tumors

A. Squamous cell carcinoma (epidermoid carcinoma)

Variant: Spindle cell carcinoma

B. Small-cell carcinoma

- Oat cell carcinoma
- Intermediate cell type
- Combined oat cell carcinomas

C. Adenocarcinoma

- Acinar adenocarcinoma
- Papillary adenocarcinoma
- Bronchio-alveolar carcinoma
- Solid carcinoma with mucus formation

D. Large-cell carcinoma

Variants: Giant cell carcinoma
Clear cell carcinoma

E. Adenosquamous carcinoma

F. Carcinoid tumor

G. Bronchial gland carcinomas

- Adenoid cystic carcinoma
- Mucoepidermoid carcinoma

II. Malignant mesothelial tumors

Malignant mesothelioma

- Epithelial
- Fibrous (spindle cell)
- Biphasic

III. Miscellaneous malignant tumors

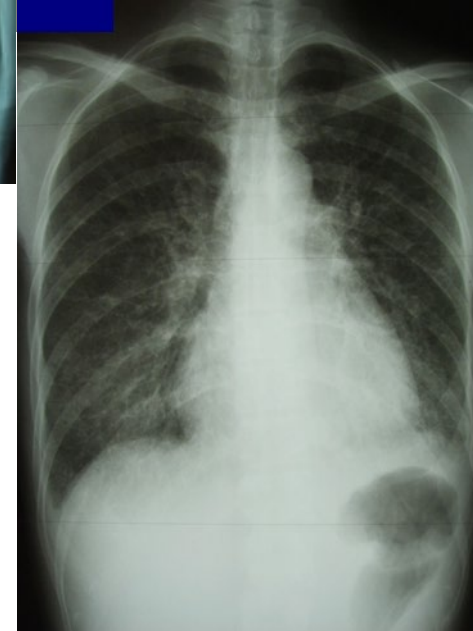
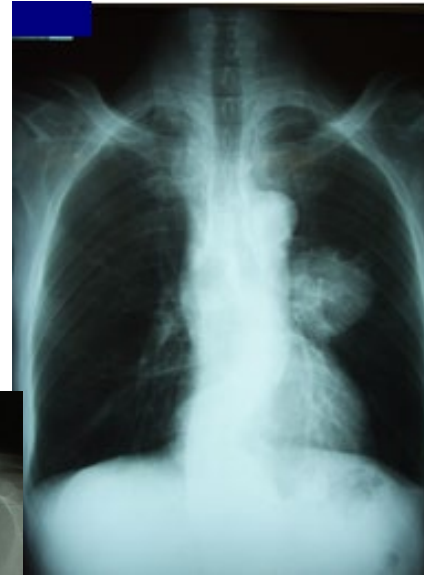
A. Carcinosarcoma

B. Pulmonary blastoma

C. Malignant melanoma

D. Malignant lymphoma

Suspected lung cancer





Different ways of recognizing lung cancer, different clinical presentation

- **Clinical symptoms and signs**
- **Clinical syndrome**
- **Incidental findings on other medical examination**
e.g. incidental findings on CT coronary angiogram or CT abdomen
- **Identification of at risk population (lung cancer screening)**



BRONCHOGENIC CARCINOMA: SYMPTOMS

Symptoms

- | | |
|--|--|
| 1. Radiological abnormality only (5%) | Nil |
| 2. Constitutional Symptoms | Malaise, ↓ appetite
and body weight |
| 3. Due to primary lesion | |
| 1. Bronchial mucosa ulceration → | Cough, sputum,
haemoptysis |
| 2. Obstructive . partial → | Wheeze, unresolved
pneumonia |
| . complete →
lung collapse | Dyspnoea |
-

BRONCHOGENIC CARCINOMA: SYMPTOMS

Symptoms

4. Due to intrathoracic spread (*relating symptoms to anatomy*)

1. spread along lymphatics to both lungs → lymphangitis carcinomatosa

Cough, dyspnoea

2. Pleura, pleural effusion

Chest pain, dyspnoea

3. Pericardial effusion → cardiac tamponade

Dyspnoea

4. Superior vena cava obstruction

Dyspnoea, stridor,
dysphagia, face swelling

5. (L) recurrent laryngeal N.

Hoarseness of voice

6. Brachial plexus (C8 T1 T2) and inferior cervical sympathetic ganglion (Horner)

Pain in shoulder and arm
loss of sweating on 1
side of face

7. Oesophagus

Dysphagia

8. Chest wall and ribs

Chest pain, swelling



Extra-thoracic manifestation

- **Cachexia**
- **Clubbing**
- **Supraclavicular lymph nodes and cervical lymph nodes enlargement**
- **Hepatic metastasis (hepatomegaly, deranged liver function)**
- **Brain metastasis (seizures, change in personality, vomiting)**
- **Bone metastasis (pathological fractures, hypercalcemia, bone pain)**
- **Adrenal metastasis (rarely cortisol insufficiency)**
- **Spinal cord metastasis (cord compression)**
- **Skin metastasis**
- **Choroidal metastasis (impaired visual acuity)**
- **Unexplained anemia**

BRONCHOGENIC CARCINOMA: SYMPTOMS

Symptoms

6. Systemic nonmetastatic syndrome

1. Connective tissue - clubbing,
hypertrophic pulmonary
osteoarthropathy

Arthralgia, pain and tenderness
of extremities

2. Ectopic hormones:

- ADH (→ ↓ Na⁺)
- ACTH (→ ↓ K⁺)
- PLP (→ ↑ Ca⁺⁺)

Confusion, weakness

Weakness

Polyuria, thirst, confusion

3. Neuromuscular

- Encephalopathy
- Cerebellar degeneration
- Peripheral neuropathy
- Myasthenia-like (Eaton-Lambert)
- Dermatomyositis

Dementia, confusion

Ataxia and clumsiness

Paraesthesia and weakness

INVESTIGATIONS

- 1. To confirm the diagnosis + histological subtypes**
 - 2. To stage the tumor + assess fitness of patient**
- to select treatment modality**

INVESTIGATIONS

1. Diagnosis

- Chest X-ray
 - Cytology
 - . sputum
 - . bronchial washing and brushings
 - . pleural fluid
 - . fine needle aspiration
 - Histology
 - . bronchoscopic biopsy
 - . bronchial
 - . transbronchial
 - . pleura
 - . lymph node
 - . other metastatic site, e.g. skin, liver
 - . exploratory thoracotomy
-

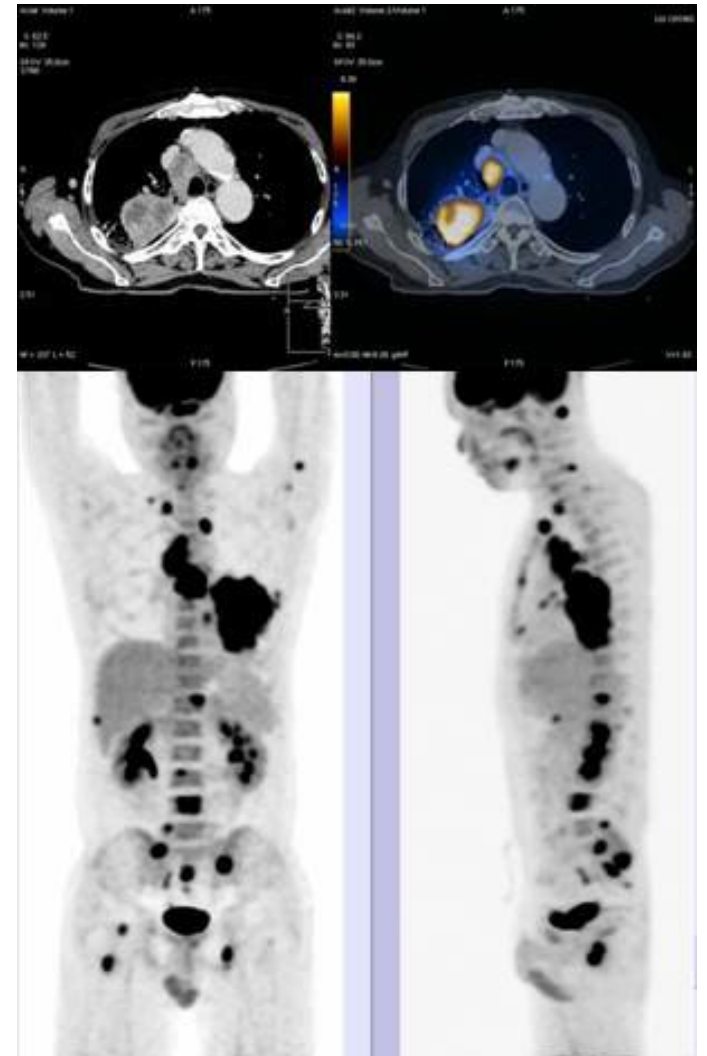
INVESTIGATIONS

2. Staging + Surgical Assessment

- blood tests . LFT, RFT, Ca⁺⁺
 - bronchoscopy
 - pleural/LN etc biopsy
 - CT thorax + upper abdomen
 - mediastinoscopy
 - CT brain) as indicated
 - bone scan)
 - cardiac assessment
 - lung function assessment
 - PET scan
-

PET-CT

- **Excellent tumor imaging**
- **FDG is taken up by cells in glycolysis but is bound within these cells and cannot enter the normal glycolytic pathway.**
- **Increased activity is demonstrated in cells with high metabolic rates, as is seen in tumors and areas of inflammation.**



Pulmonary Interventions

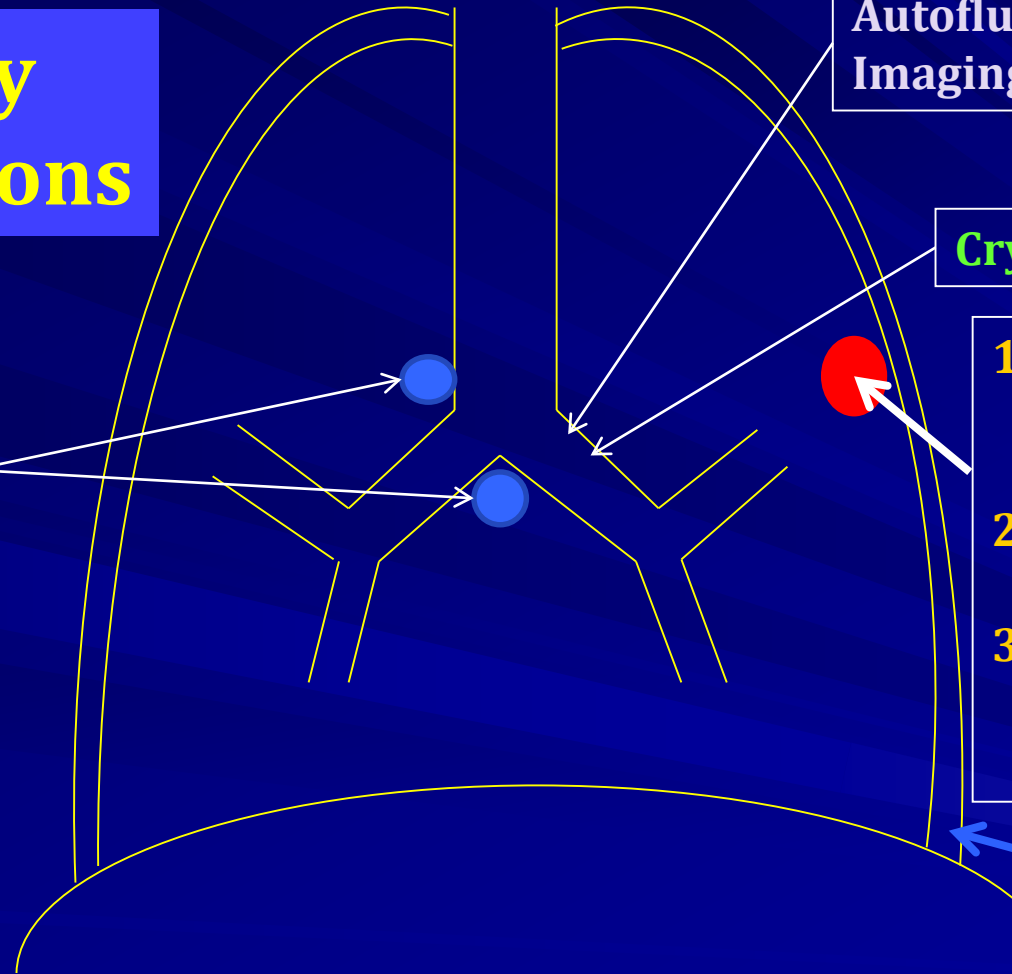
Autofluorescence
Imaging

Cryotherapy

1. Radial probe (RP)-EBUS with Guided sheath
2. Navigational bronchoscopy
3. Image-guided percutaneous biopsy

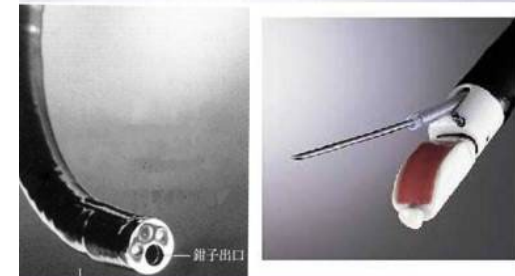
Pleuroscopy/
Indwelling pleural
catheter

Convex probe (CP)-
EBUS-TBNA



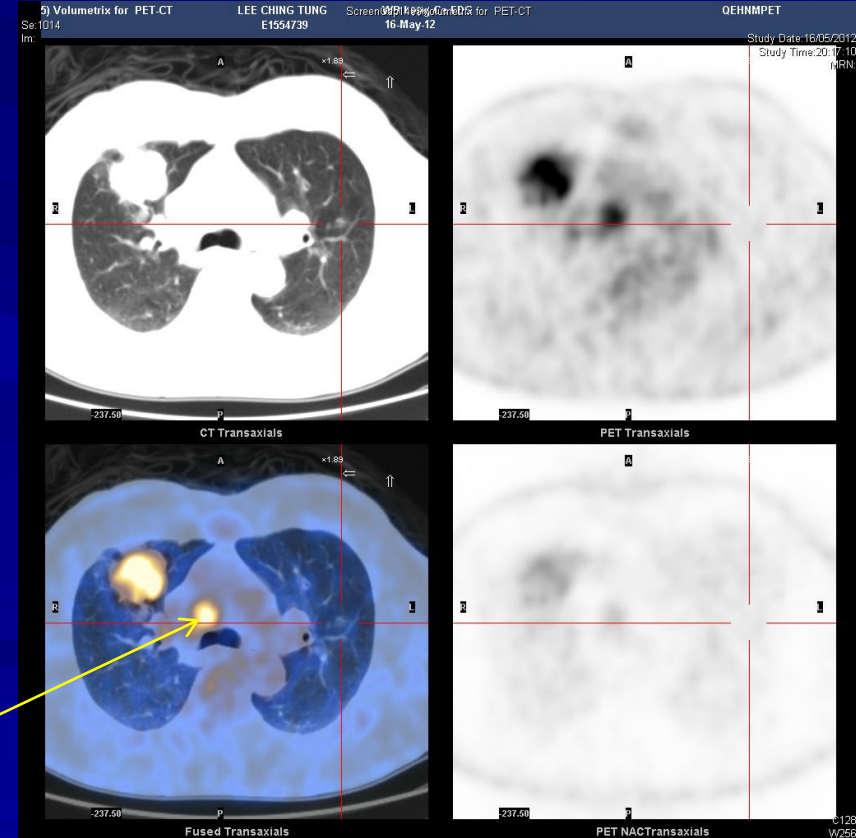
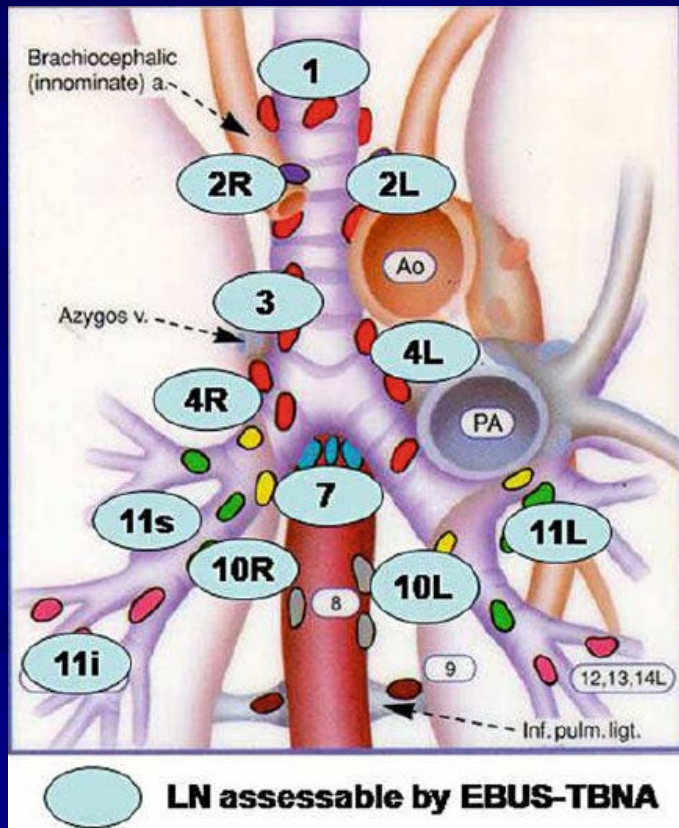
Endobronchial ultrasound (EBUS)

- Can assess subcarinal, upper and lower paratracheal and hilar lymph nodes
- Sensitivity ~ 90%
- False Negative rate ~ 20%



Endobronchial Ultrasound Transbronchial Needle Aspiration (EBUS-TBNA)

(Convex probe)



Indications for CP-EBUS-TBNA

- Staging for lung cancer

 - Sampling of mediastinal LN

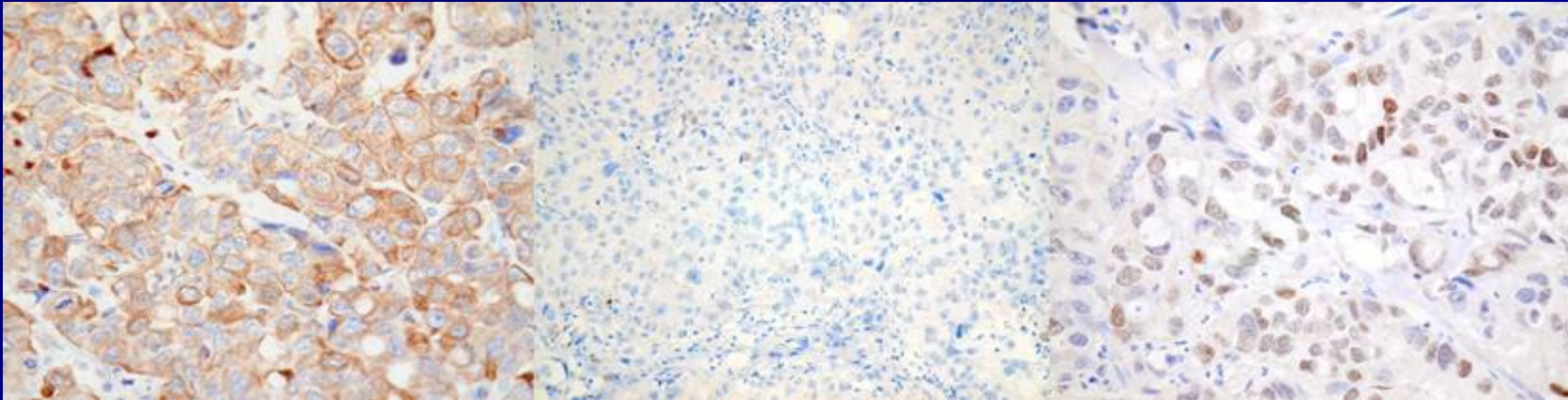
- Alternative diagnosis for mediastinal lymphadenopathy

 - Tuberculosis, sarcoidosis, lymphoma

- Tumor mass immediately adjacent to accessible part of the airway through the EBUS bronchoscope

Endobronchial biopsy

Endobronchial biopsy showed adenocarcinoma
(CK7 positive, CK20 negative and TTF1 positive)



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Pleuroscopy (Medical Thoracoscopy): Semi-rigid instrument



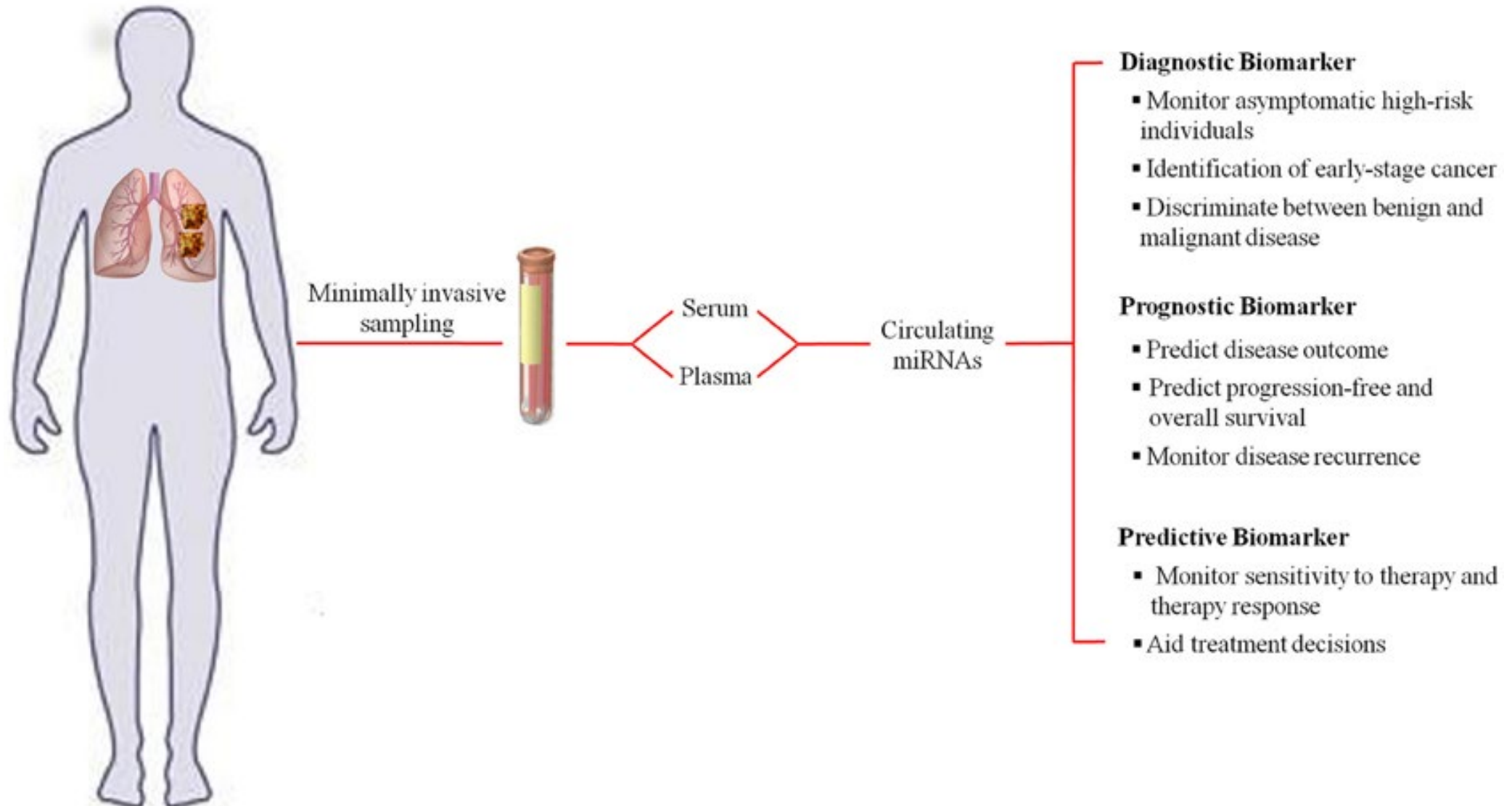
- Handling ~ flexible bronchoscope
- 1 port of entry
- flexible trocar



Distal 5 cm: 160° up/130° down
Outer diameter: 7mm
Working channel: 2.8mm



Liquid biopsy – to supplement when tissue biopsy difficult or impossible





Liquid biopsy – to supplement when tissue biopsy difficult or impossible

- **Body fluids (plasma, serum, bronchoalveolar lavage, urine)**
- **Gold-standard for diagnosis is still lung cancer tissue for histological diagnosis**

Clinical applications of circulating biomarkers

- **For diagnosis**
- **To guide targeted therapy selection**
- **To monitor disease progress**
- **To monitor treatment response**
- **To identify new targets**
- **Early detection of disease recurrence**
- **For prognostication**

Examples: CTC, ctDNA, miRNA, tumor derived exosomes or oncogenic proteins	Primary tumor histology, metastatic site biopsy
Relatively non-invasive	Mostly invasive intervention or surgery
Allows serial monitoring	At diagnosis +/- re-biopsy on disease progression/drug failure
Lack of a standardized techniques, especially with amplification of minute amount of materials	Well-established techniques or even Gold standard tests
Tumor cell material leakage dependent: variable test sensitivity	Sampling representativeness of small biopsy

Staging for Non-Small Cell Lung Cancer

Based on the TNM Staging Classification

Tumour size and local invasion

**N Lymph nodes -- hilar, mediastinal or extra-thoracic
-- ipsilateral vs contralateral**

M Metastasis

Stages I -- early disease

II -- early disease

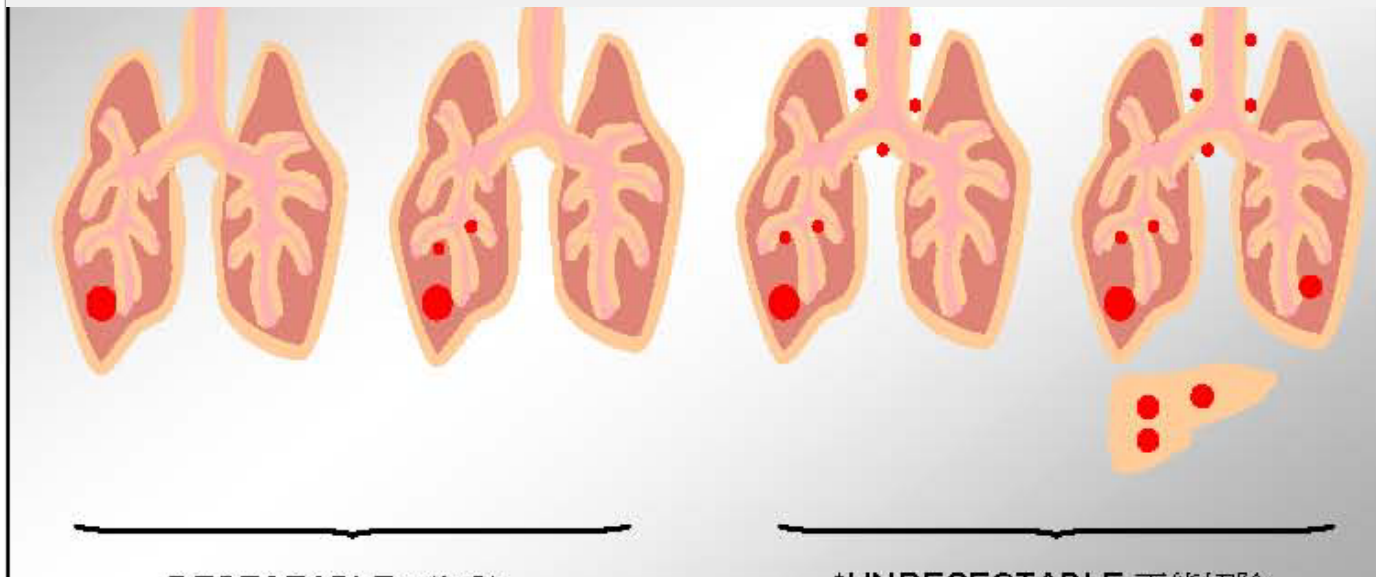
III -- locally advanced disease

IV -- advanced or metastatic disease

NB: Updated 8th ed. TNM staging manual in 2018

Lung cancer staging 肺癌分期

• EBUS-TBNA 超聲波針吸活檢



early

early

locally advanced

metastatic

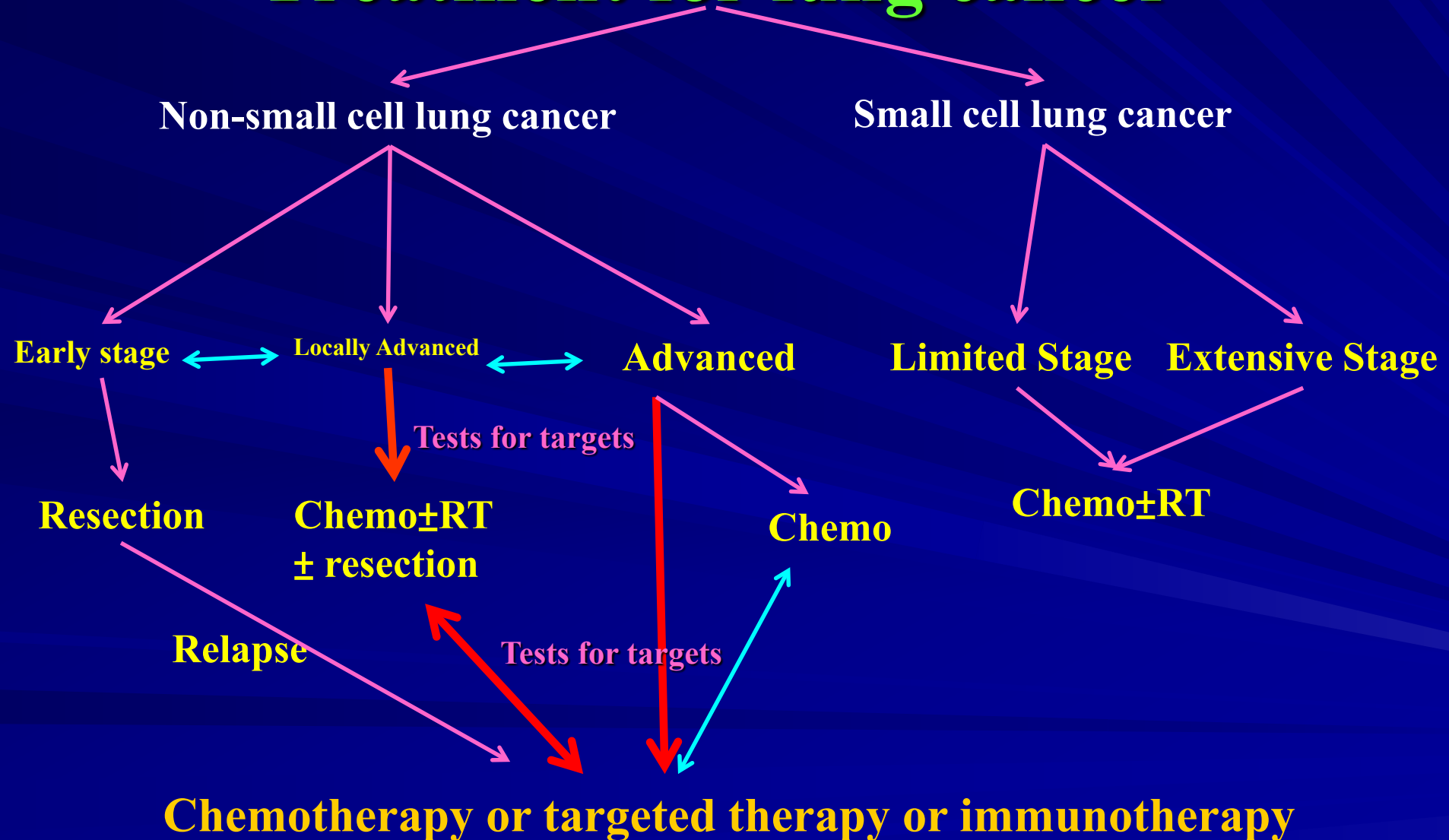
← OPERABLE →

← INOPERABLE →

SURGERY SURGERY
(if medically fit)

RT + CT CT
(if good performance status)

Treatment for lung cancer



Chemotherapy for Advanced stage NSCLC

Cyclical combination chemotherapy

A platinum drug (e.g. *cisplatin* or *carboplatin*)

+

**A newer generation cytotoxic drug
(e.g. *Pemetrexed*, *Taxol*, *Gemcitabine*)**

Response rate about 30 - 40%

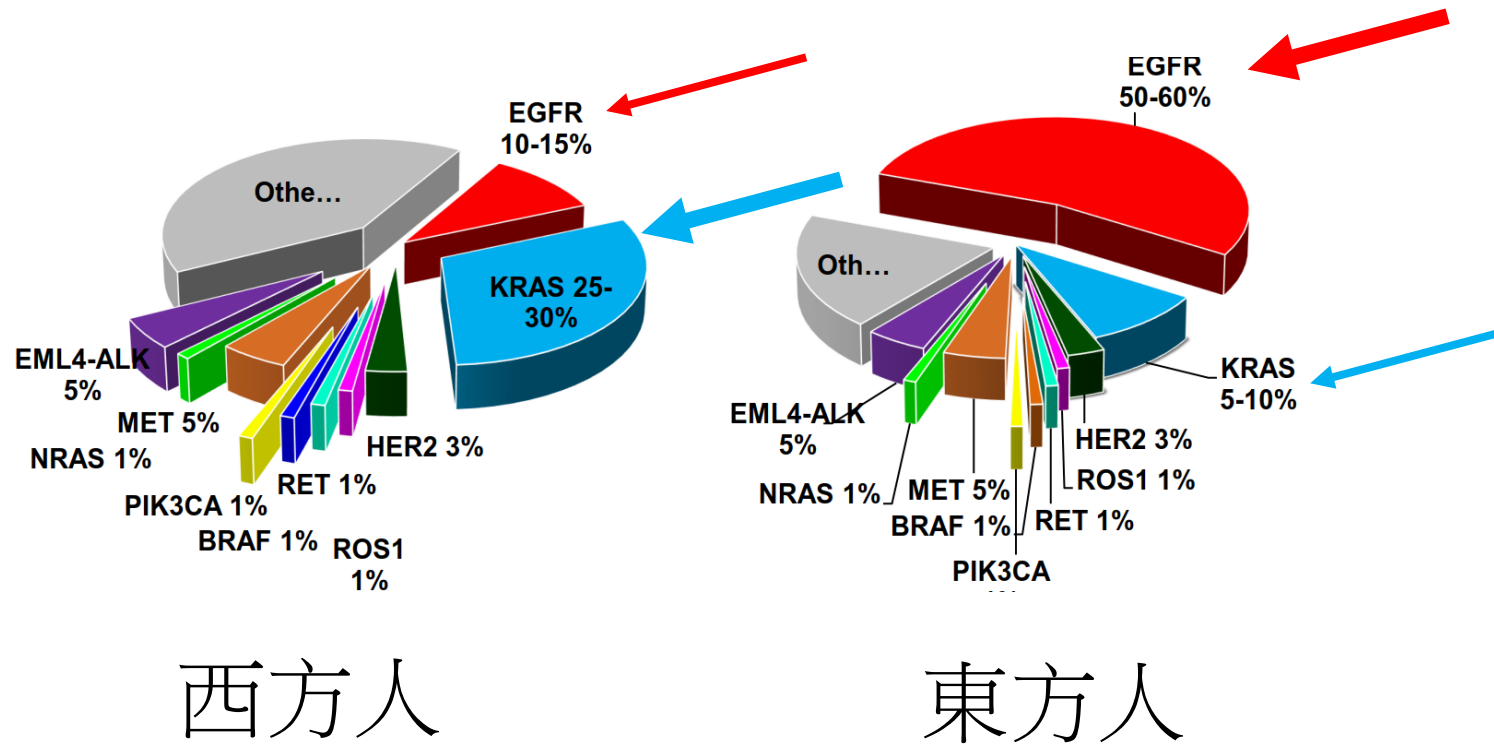
Note side effects especially myelosuppression



Workup for lung cancer systemic treatment - especially cytotoxic chemotherapy

- Check hepatitis B carrier status
If hepatitis B carrier, cover with appropriate antiviral prophylaxis e.g. entecavir, throughout course of chemotherapy
- Premedications for some chemotherapy regimes:
e.g. Vitamin B₁₂ and folic acid supplementation to prepare for chemotherapy with Pemetrexed.

東西方人的肺癌基因不同



Courtesy of Prof PC Yang, National Taiwan Univ.

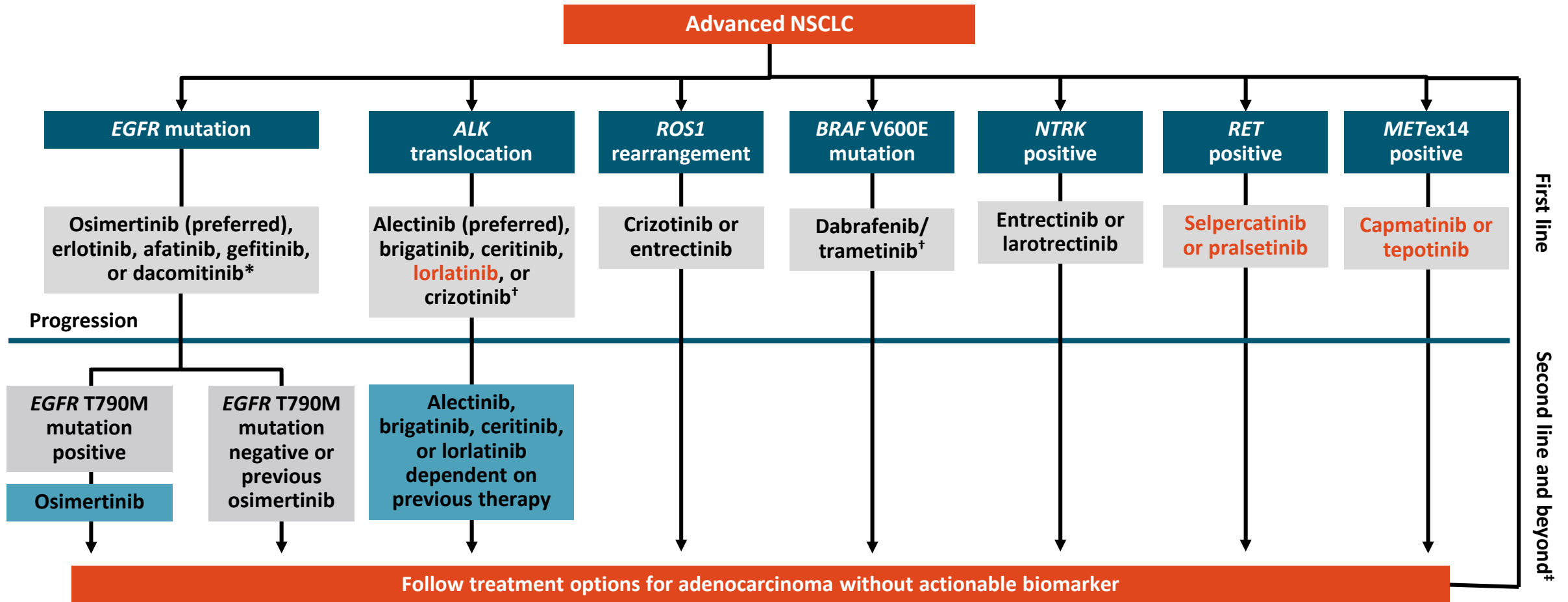


Molecular Targeted Therapy

- **Indications**

As primary treatment for tumor with oncogenic molecular targets

Treatment for advanced stage NSCLC with actionable mutations



Epidermal growth factor receptor (*EGFR*) Tyrosine Kinase inhibitors (TKI)

- Biomarkers: Sensitizing *EGFR* mutations (15-nucleotide deletion in exon 19, L858R or L861Q in exon 21)
- Resistant T790M mutation in exon 20
(Mutations detectable in tumors or plasma DNA)

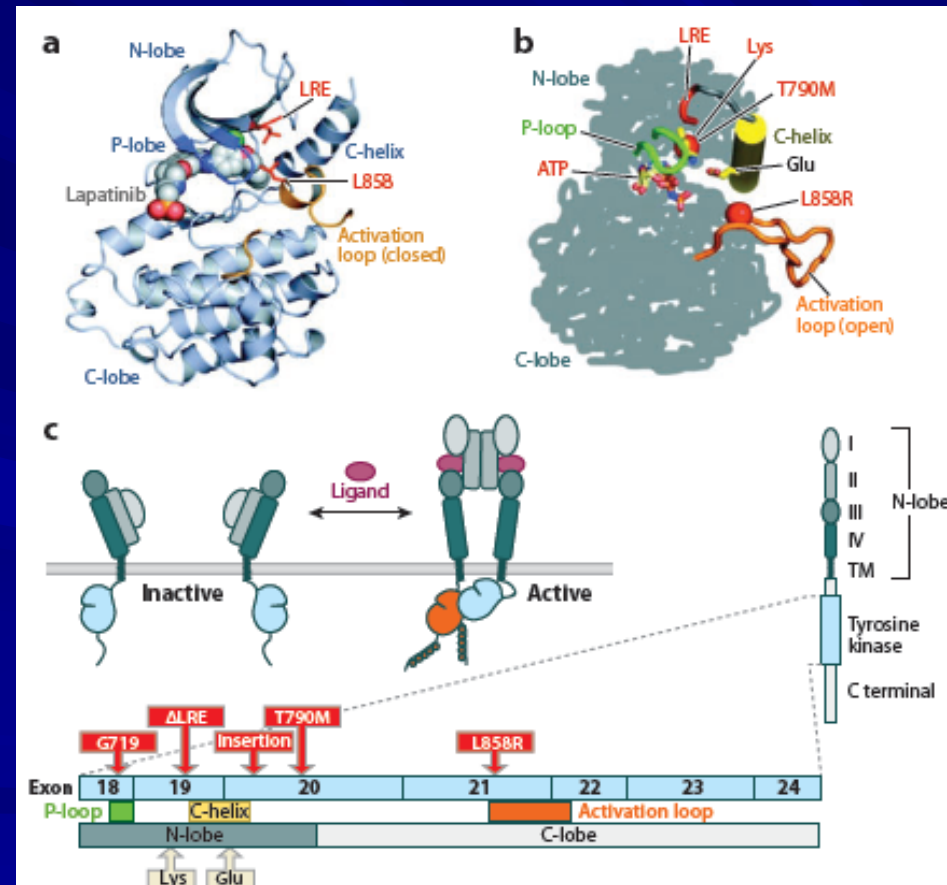
Clinical indications:

Sensitizing *EGFR* mutations

-> Gefitinib/Erlotinib/Afatinib

Resistant *EGFR* mutations

-> Osimertinib



G Santos, Ann Rev Patho 2010

Common adverse effects from EGFR –TKI and some other TKIs

WHO Grade III Stomatitis/Mucositis (can only tolerate fluid diet)



Paronychia with ingrown toe nail

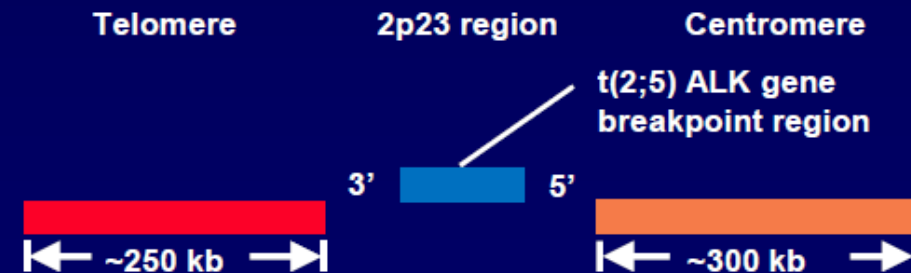
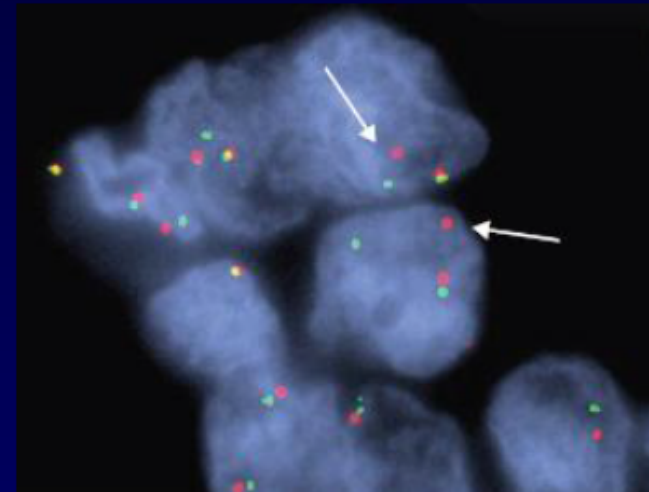


ALK-Positive NSCLC

ALK translocation in NSCLC → oncogenic activation

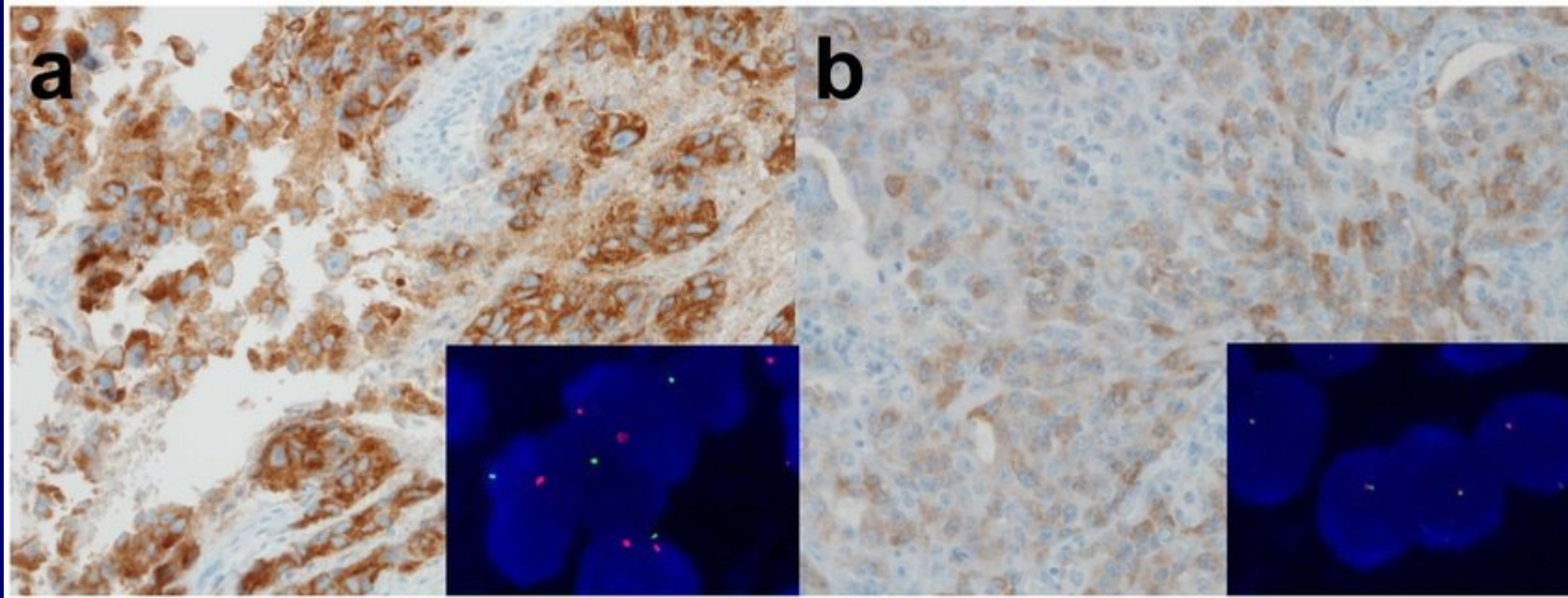
- Adenocarcinomas
- Never or light smokers
- Median age 50 years
- No sex differences

Break-apart FISH assay
for ALK-fusion genes



Anaplastic Lymphoma Kinase (*ALK*) rearrangement

■ **Biomarkers:** *ALK* immunohistochemical staining
or *ALK* FISH



Clinical indications: *ALK* re-arranged or IHC +ve tumor

First line: Crizotinib

Second line or beyond: Ceritinib/Alectinib/Brigatinib/Lorlatinib
or systemic chemo/immunotherapy

Immunotherapy: Immune-checkpoint inhibitor

■ PDL1/PD1 axis

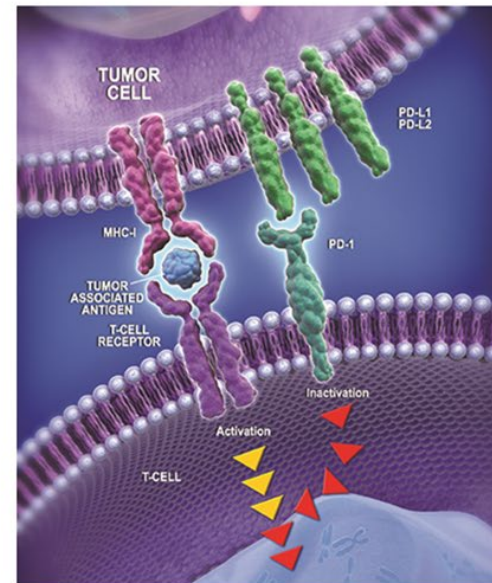
■ Monoclonal antibodies that target either

1. Programmed Death-ligand 1 expression on tumor cells, or

2. Programmed Death 1 expression on cytotoxic T cells

■ E.g: Pembrolizumab, Nivolumab, Atezolizumab

- Potent, humanized monoclonal antibody against PD-1 that prevents PD-1 from binding to its ligands, PD-L1 and PD-L2
- Robust antitumor activity and manageable toxicity in multiple tumor types
- As monotherapy for previously treated and treatment-naïve advanced NSCLC in KEYNOTE-001 (n = 495)¹
 - 19.4% ORR, 3.7-month median PFS, 12.0-month median OS
 - 45.2% ORR in patients with PD-L1 expression in ≥50% of tumor cells



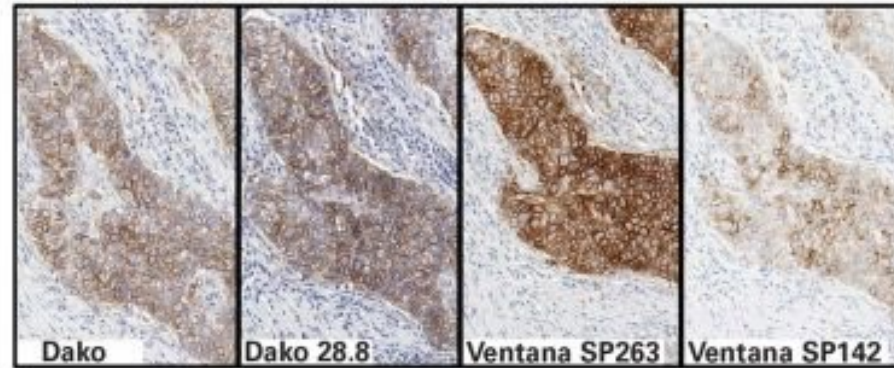
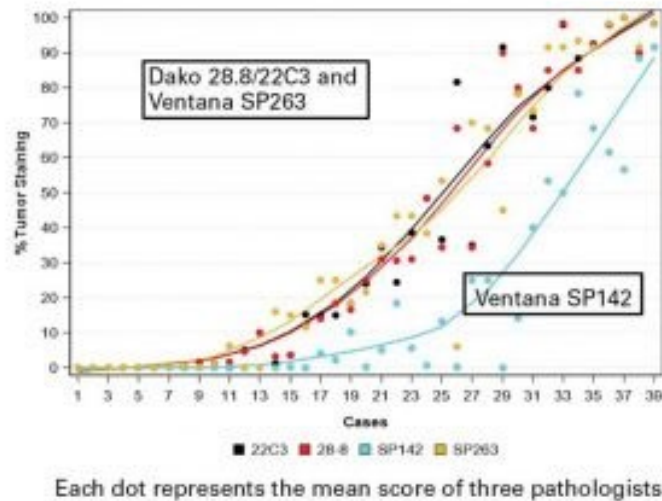
Anti-PDL1 therapy

Biomarkers:



PDL1 immunohistochemical staining intensity

Tumor proportional score (TPS) > 50% predicts better response to anti-PDL1 therapy



Conclusion: 1: Three assays showed similar staining characteristics for PD-L1 staining on tumor cells, but Ventana SP142 comparatively showed fewer tumor cells stained.

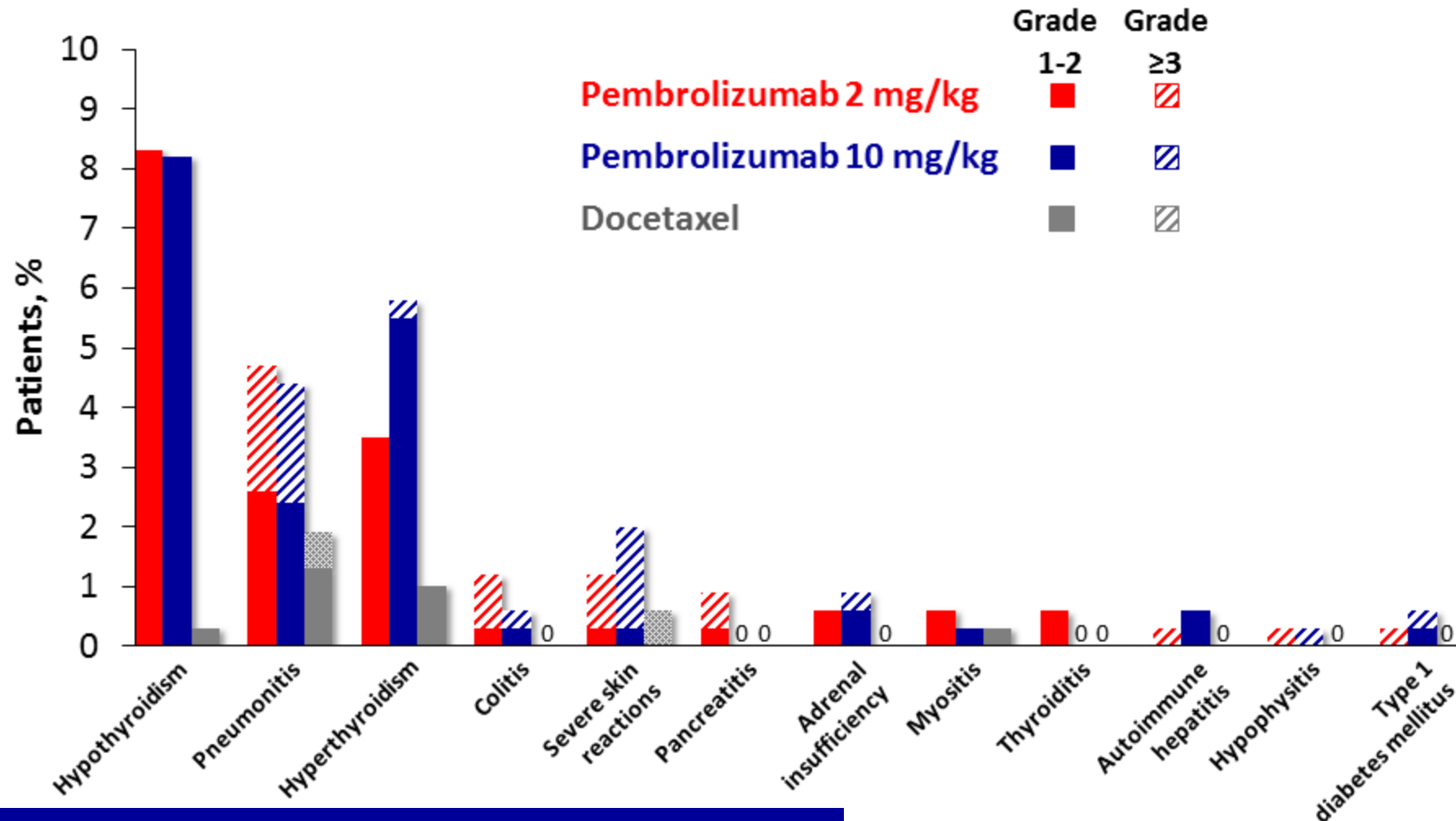
Clinical indications:

1st line treatment: anti-PDL1 therapy combined with chemotherapy in tumors with PDL1 expression

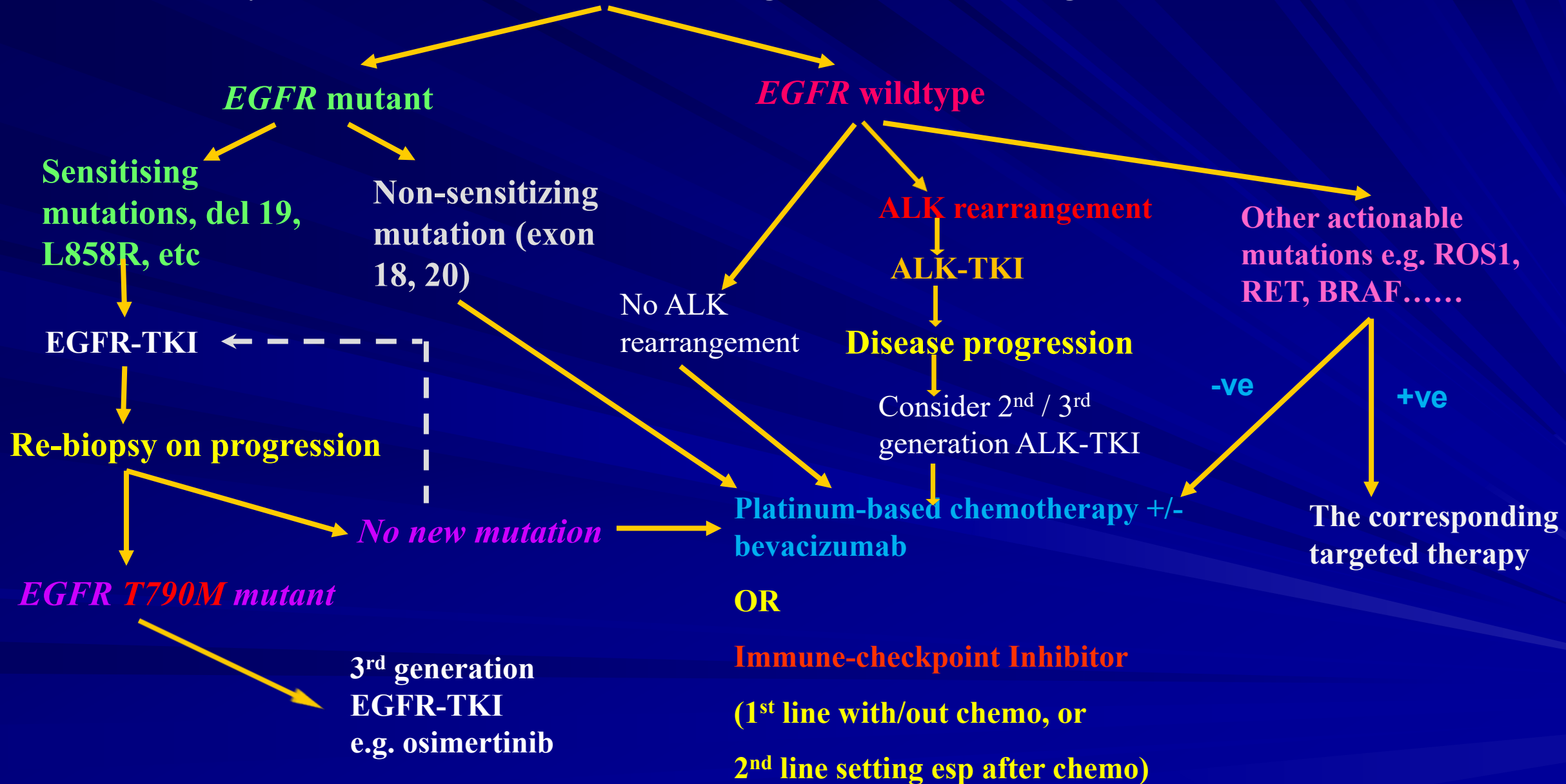
2nd line or beyond treatment: anti-PDL1 therapy alone (no need for PDL1 test)

F Hirsch, J Thor Oncol. 2017;12(2) 208-222

Immune-Mediated AEs Occurring in ≥ 2 Patients in the Pembrolizumab Arms



Systemic treatment for advanced stage non-small cell lung cancer



MANAGEMENT

4. Supportive

- Hospice care
 - Analgesics + other symptomatic care, e.g. cough suppression, thoracentesis + pleurodesis
 - Psychosocial support
-

Summary

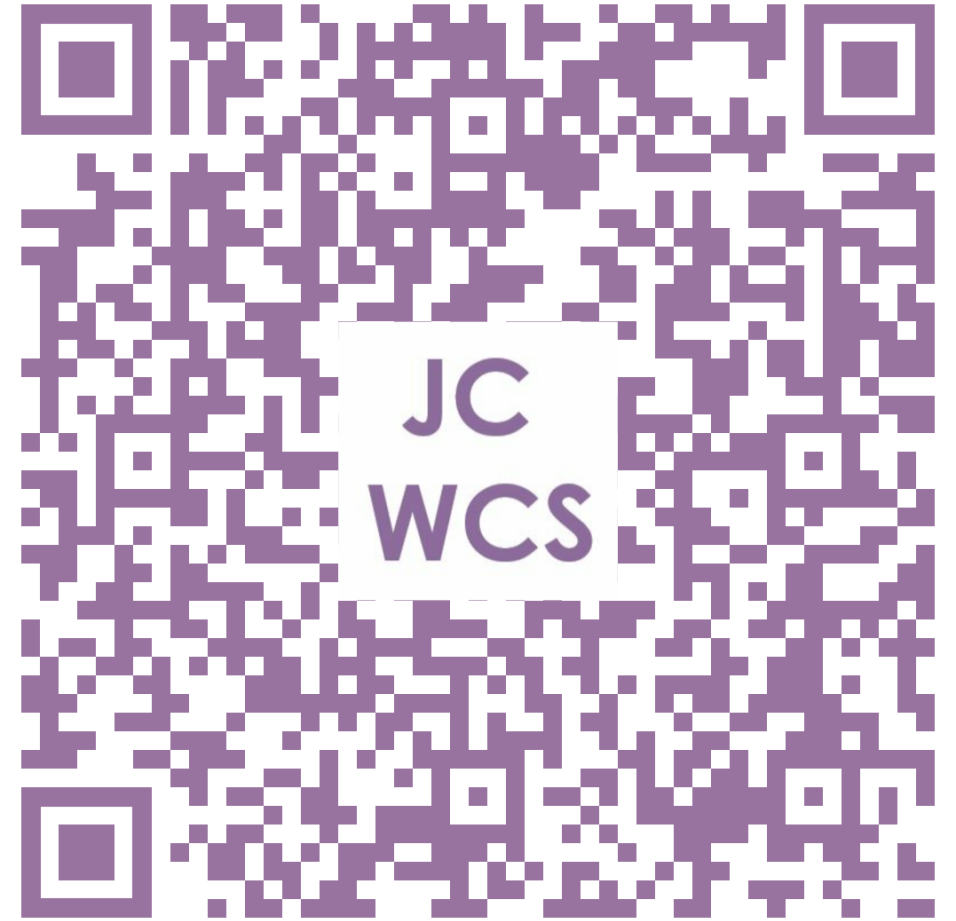
- **Variable clinical presentation of lung cancer, from asymptomatic to frank systemic disturbance or clinical syndromes**
- **Different diagnostic methods to suit individual presentation**
- **The aims of diagnostic investigation should be:**
 1. **distinction between benign and malignant lesion**
 2. **to obtain tissue diagnosis, especially for at risk population**
 3. **to work up for staging with diagnosis of lung cancer and to evaluate for metastasis**
 4. **functional assessment for patients in preparation for treatment modalities**

TELL US WHAT

JC Whole Class Session

YOU THINK

THANK YOU!



Look for today's teacher
and the others you may have missed.
(HKU Portal login required)